

RFID Testing Challenges for Complex RF Environments

A brief overview of RFID technology and testing challenges for complex operating environments where throughput and communication issues may be caused by multiple readers, dense-mode environments and pre-existing non-RFID signals

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The applications of radio-frequency identification (RFID) are growing rapidly as equipment prices drop and global markets expand. The use of embedded RFID is increasing. Coordinating bodies such as the Ubiquitous ID Center and the T-Engine Forum have been formed. And the GSM Association now supports embedding of RFID-based near-field communications capability into cellular phones.

However, a big challenge in RFID is optimising throughput (data reading

RFID systems may need to function, when integrated into the same device, with cellular, WLAN, Bluetooth or Zigbee technologies. Finally, interference from other users in the same band needs to be considered.

The result is a need to simulate complex RF environments and analyse the performance of the RFID system under these conditions before deployment. The pulsed nature of RFID and the typical interference sources make this task all the more difficult.

RFID technology overview

An RFID system at its simplest consists of a tag, which may be passive, and a reader. Architecturally, reading passive tags is somewhat different from the traditional full-duplex data link. Unlike traditional active data links, passive tags rely on the received RF energy to power themselves. These also do not generate their own transmit carrier signal.

Rather, they modulate some of the energy being transmitted by the interrogator to the tag in a process known as backscattering.

By changing the loading of the tag's antenna from absorptive to reflective, a continuous-wave (CW) signal from the interrogator can be modulated. This process is very similar to using a mirror and the sun to signal someone at a distance. It also eliminates the need for precision frequency sources and power-hungry transmitters in the tag. Since

the reader and the tag share the same frequency, these must take turns sending information. Backscattering restricts communications between the reader and the tag to a half-duplex system.

Since the uplink from tag T to reader R (denoted by T→R) is modulated from the interrogator's CW signal, it is possible to use spread-spectrum techniques such as frequency hopping. Any spreading or hopping on the interrogator's signal is automatically removed in the homodyne down-conversion of the receiver, as it shares the same local oscillator signal.

This simple system turns complex when multiple tags, multiple readers and interference are present. Let's take a look at two RFID design challenges that come from these deployment issues.

Multiple-reader and dense-mode environments

The broadband nature of passive RFID tag presents some challenges for dense (multiple) reader sites. Since the tag reader sets the system's frequency of operation, and the tag is a broadband device that responds to any reader, the tag has limited ability to respond to a specific reader. Passive tags may try to respond to all readers that are interrogating them.

Many RFID systems are implemented in a multiple-reader or dense-mode environment. Here are some definitions:

1. Single-reader environment: A



speed) in complex, or even harsh, RF environments. Passive RFID tags may respond to any reader or readers in range. Protocols exist to work with this behaviour, but the result is a complex communications behaviour that can be difficult to test without the right equipment. In addition, embedded